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INFO601 – Data Modelling & SQL

Assessment 1 – Data Modelling

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Module: Data Modelling & SQL

Assessment: Assessment 1

Database Design Report

CyberSecure Cloud Solutions

1. Introduction

CyberSecure Cloud Solutions is a cybersecurity company that provides cloud storage, virtual machines, and cybersecurity monitoring services to business clients. The company requires a database system to manage clients, accounts, services, subscriptions, billing, and cybersecurity incidents. The purpose of this database design is to create a structured and efficient data model that supports these business needs, follows database normalisation principles, and allows efficient querying and reporting. This design is based on the scenario and business rules provided in the assignment brief.

2. Identified Entities and Attributes

The following entities were identified from the business rules.

2.1 Client

Attribute	Description	Key
ClientID	Unique identifier for each client	Primary Key
Name	Client name	
ContactEmail	Client email address	
ContactPhone	Client phone number	
Address	Client address	

Explanation:

Each client has a unique ClientID, and the client details include Name, ContactEmail, ContactPhone, and Address. A client can have multiple accounts, but each account belongs to only one client.

2.2 Account

Attribute	Description	Key
AccountID	Unique identifier for each account	Primary Key
AccountName	Account name	
ClientID	References the client who owns the account	Foreign Key

Explanation:

Each account is linked to one client. The business rules state that a client can have multiple accounts, and each account is associated with only one client. The brief also mentions SubscriptionDate under accounts, but in a normalised design this is better stored in the Subscription entity so the date is not duplicated.

2.3 Service

Attribute	Description	Key
ServiceID	Unique identifier for each service	Primary Key
ServiceName	Name of the service	
ServiceDescription	Description of the service	

Explanation:

The company offers three services: Cloud Storage, Virtual Machines, and Cybersecurity Monitoring. Each service has a unique ServiceID, ServiceName, and ServiceDescription.

2.4 Subscription

Attribute	Description	Key
SubscriptionID	Unique identifier for each subscription	Primary Key
AccountID	References the subscribed account	Foreign Key
ServiceID	References the subscribed service	Foreign Key
SubscriptionDate	Date the subscription started	
SubscriptionStatus	Status of the subscription	

Explanation:

An account can subscribe to multiple services, and a service can be subscribed to by multiple accounts. This is a many-to-many relationship, so it is resolved using the Subscription entity. The brief states that each subscription includes ServiceID, AccountID, and SubscriptionDate. I added SubscriptionStatus because it is useful for identifying whether a subscription is active, paused, or cancelled.

2.5 Billing

Attribute	Description	Key
BillingID	Unique identifier for each billing record	Primary Key
SubscriptionID	References the related subscription	Foreign Key

BillingDate	Date the bill was generated	
Amount	Billing amount	
PaymentStatus	Status of payment	

Explanation:

Each subscription can have multiple billing records. Each billing record is uniquely identified by BillingID and must include SubscriptionID, BillingDate, Amount, and PaymentStatus.

2.6 CybersecurityIncident

Attribute	Description	Key
IncidentID	Unique identifier for each incident	Primary Key
AccountID	References the account involved in the incident	Foreign Key
IncidentDate	Date the incident occurred	
IncidentType	Type of incident	
IncidentStatus	Current status of the incident	
Level	Severity level of the incident	
Description	Detailed incident description	

Explanation:

Each account may have multiple cybersecurity incidents. Each incident is uniquely identified by IncidentID and includes AccountID, IncidentDate, and IncidentType. I included IncidentStatus, Level, and Description because these are useful for tracking unresolved incidents and incident severity in a real-world cybersecurity system.

3. Relationships Between the Entities

The following relationships were identified from the business rules.

Relationship	Type	Rationale
Client to Account	One-to-Many	One client can have many accounts, but each account belongs to only one client
Account to Subscription	One-to-Many	One account can subscribe to many services over time
Service to Subscription	One-to-Many	One service can be subscribed to by many accounts
Subscription to Billing	One-to-Many	One subscription can generate many billing records
Account to CybersecurityIncident	One-to-Many	One account can have many incidents

Relationship explanation:

The Client table is linked to the Account table through ClientID. The Account and Service tables are linked through the Subscription table because the business rule is many-to-many between accounts and services. Billing records are linked to Subscriptions because each subscription may create multiple invoices or payment records. Cybersecurity incidents are linked directly to Account because incidents occur at the account level. Primary keys uniquely identify each record, while foreign keys preserve referential integrity across tables.

4. Entity-Relationship Diagram (ERD)

The ERD for this database should show the following entities and links:

- Client
- Account
- Service
- Subscription
- Billing
- CybersecurityIncident

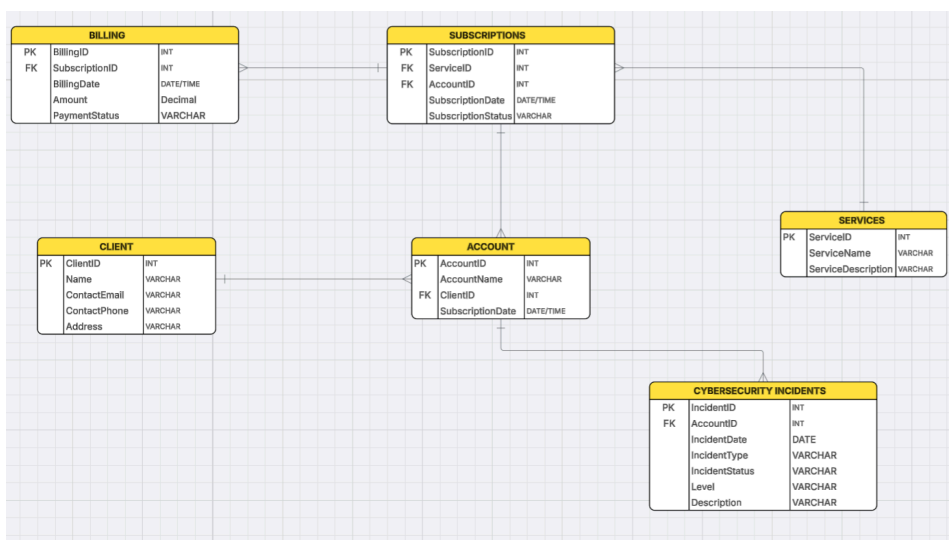
ERD relationships to include:

- Client 1:M Account
- Account 1:M Subscription
- Service 1:M Subscription
- Subscription 1:M Billing
- Account 1:M CybersecurityIncident

ERD notes:

- Mark primary keys clearly.
- Mark foreign keys clearly.
- Use crow's foot notation for one-to-many relationships.
- Show Subscription as the associative entity between Account and Service.

ERD placeholder:



The assignment requires the ERD to include all entities, attributes, and relationships, and the marking schedule gives significant weight to clarity and completeness of the ERD.

5. Normalisation

The database design follows normalisation up to Third Normal Form (3NF).

First Normal Form (1NF)

The design satisfies 1NF because all attributes are atomic and there are no repeating groups. Each field contains a single value, such as one email address, one phone number, or one billing amount.

Second Normal Form (2NF)

The design satisfies 2NF because all non-key attributes depend on the full primary key of their table. For example, in the Subscription table, SubscriptionDate and SubscriptionStatus depend on SubscriptionID, not on only part of a composite key. The many-to-many relationship between Account and Service is separated into its own table, which prevents partial dependency problems.

Third Normal Form (3NF)

The design satisfies 3NF because there are no transitive dependencies. Non-key attributes depend only on the primary key, and related data is stored in separate tables. For example, billing data is stored in Billing instead of being repeated in Subscription, and incident details are stored in CybersecurityIncident instead of being duplicated elsewhere. This reduces redundancy and avoids update anomalies.

The assignment specifically requires normalisation to 3NF, and the marking rubric gives separate marks for correct application of 1NF, 2NF, and 3NF.

6. Optimisation and Indexing Strategies

The assignment asks for indexing strategies and other methods to improve query performance, especially for active subscriptions, billing reports, and unresolved incidents. It also recommends indexing commonly queried fields such as ClientID, ServiceID, and IncidentDate.

6.1 Recommended Indexes

Table	Field(s) to Index	Reason
Client	ClientID	Fast lookup of client records
Account	ClientID	Faster join between Client and Account
Subscription	AccountID, ServiceID	Faster subscription lookups and joins
Billing	SubscriptionID, BillingDate	Faster billing report generation
CybersecurityIncident	AccountID, IncidentDate, IncidentStatus	Faster incident tracking and filtering unresolved incidents

Service	ServiceID	Faster service lookup
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6.2 Composite Indexes

Table	Composite Index	Purpose
Subscription	(AccountID, ServiceID)	Improves searches for subscriptions by account and service
CybersecurityIncident	(IncidentStatus, IncidentDate)	Improves queries for unresolved or recent incidents
Billing	(SubscriptionID, BillingDate)	Improves invoice history searches

6.3 Other Optimisation Strategies

Strategy	Explanation
Use foreign key constraints	Maintains data integrity and improves relationship consistency
Store only necessary data	Reduces duplication and improves storage efficiency
Use efficient datatypes	Saves space and supports faster processing
Separate transactional and reporting queries	Improves performance for operational use
Use proper validation in forms	Prevents invalid data entry at the source

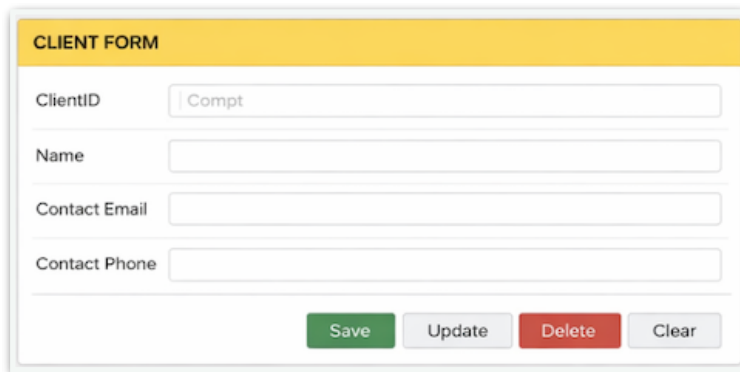
These strategies support efficient retrieval of active subscriptions, billing reports, and unresolved incidents, which are specifically mentioned in the brief.

7. Draft Data Entry Forms

The assignment requires draft forms for the main entities, and the forms must capture all necessary attributes while validating relationships between entities.

7.1 Client Form

Field	Input Type	Validation
ClientID	Auto-generated / text	Must be unique
Name	Text box	Required
ContactEmail	Text box	Must be a valid email format
ContactPhone	Text box	Must be a valid phone number
Address	Text area	Required

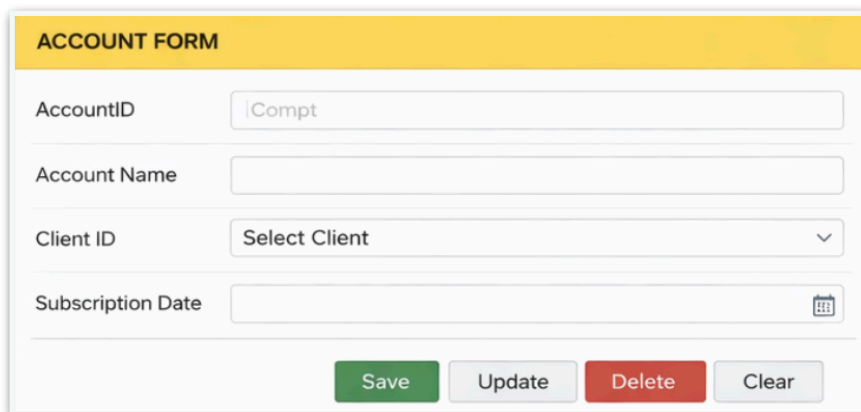


Purpose:

Used to store client details.

7.2 Account Form

Field	Input Type	Validation
AccountID	Auto-generated / text	Must be unique
AccountName	Text box	Required
ClientID	Drop-down list	Must select an existing client



Purpose:

Used to create an account linked to a specific client.

7.3 Service Form

Field	Input Type	Validation
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ServiceID	Auto-generated / text	Must be unique
ServiceName	Text box	Required
ServiceDescription	Text area	Required

SERVICE FORM

ServiceID

Compt

Service Name

Service Description

Save

Update

Delete

Clear

Purpose:

Used to store the available services offered by the company.

7.4 Subscription Form

Field	Input Type	Validation
SubscriptionID	Auto-generated / text	Must be unique
AccountID	Drop-down list	Must select an existing account
ServiceID	Drop-down list	Must select an existing service
SubscriptionDate	Date picker	Required
SubscriptionStatus	Drop-down list	Active, Suspended, Cancelled

SUBSCRIPTION FORM

Subscription ID

Select Service

▼

Service ID

Select Account

▼

Subscription Date

Active

📅

Save

Update

Cancel Subscription

Clear

Purpose:

Used to register which service an account has subscribed to.

7.5 Billing Form

Field	Input Type	Validation
BillingID	Auto-generated / text	Must be unique
SubscriptionID	Drop-down list	Must select an existing subscription
BillingDate	Date picker	Required
Amount	Number / currency	Must be greater than zero
PaymentStatus	Drop-down list	Paid, Unpaid, Pending

BILLING FORM

Billing ID

Compt

Subscription ID

Select Subscription

▼

Billing Date

📅

Amount

Payment Status

Paid

▼

Save

Update

Generate Invoice

Clear

Clear

Purpose:

Used to record billing information for subscriptions.

7.6 Cybersecurity Incident Form

Field	Input Type	Validation
IncidentID	Auto-generated / text	Must be unique
AccountID	Drop-down list	Must select an existing account
IncidentDate	Date picker	Required
IncidentType	Drop-down list / text	Required
IncidentStatus	Drop-down list	Open, Investigating, Resolved
Level	Drop-down list	Low, Medium, High, Critical
Description	Text area	Required

CYBERSECURITY INCIDENT FORM

Incident ID

Compt

Account ID

Select Account

Incident Date

Select Type

Incident Status

Open

Level

Low

Save

Update

Close Incident

Clear

Purpose:

Used to record and monitor cybersecurity incidents related to accounts.

8. Conclusion

This database design provides a complete and normalised solution for CyberSecure Cloud Solutions. The entities, attributes, and relationships directly reflect the business rules in the assignment brief. The design supports client management, account management, service subscriptions, billing, and incident tracking. It also follows 3NF, includes indexing strategies for performance, and provides draft data entry forms for the main entities. Overall, this structure is efficient, scalable, and suitable for the company's operational needs.